

5 Laryngology and the Upper Aerodigestive Tract

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5.0 Anatomy and Physiology of the Upper Aerodigestive Tract

The upper aerodigestive tract is composed of the nasal cavity, nasopharynx, oral cavity, oropharynx, hypopharynx, larynx, trachea, and esophagus. The complex anatomy and physiology supports basic functions in respiration, phonation, deglutition, and the special sense apparatus for the olfactory and gustatory systems. Important aspects of anatomy and physiology are reviewed here. Nasal and paranasal sinus anatomy and physiology are covered in **Chapter 4.0**.

◆ Oral Cavity

General

The vestibule includes the mucosal surface of the lips, buccal mucosa, and buccal/lateral surfaces of the alveolar ridges. The remainder of the oral cavity includes the more medial structures, including the hard and soft palate, mobile tongue (anterior two-thirds), and the oral floor. The oral floor contains the sublingual salivary glands, and the openings of the submandibular ducts (Wharton's ducts), draining the submandibular glands, are found throughout on either side of the midline. The frenulum attaches the anterior tongue to the midline oral floor. Minor salivary glands coat the oral cavity and pharynx.

Musculature

The vestibule includes the orbicularis oris, various levators and depressors, as well as the buccinator, all muscles of facial expression. Tongue musculature involves both intrinsic muscles and extrinsic muscles, including the genioglossus, hyoglossus, and styloglossus, all of which are innervated by the hypoglossal nerve.

Blood Supply

The lingual artery is the primary blood supply to the tongue. The facial artery, a branch of the external carotid, supplies the vestibule via the superior and inferior labial arteries. The greater and lesser palatine foramina in the lateral hard palate house the greater and lesser palatine arteries, branches of the maxillary artery.

Lymphatic Drainage

Facial lymphatics drain primarily to submental, submandibular, and facial nodes of level 1, while the anterior tongue lymphatics drain to upper jugular nodes of level 2, often bilaterally.

Nerve Supply

The hypoglossal nerve, cranial nerve (CN) XII, is the motor supply to the tongue. The lingual nerve (a branch of the mandibular nerve, CN V₃) provides

sensation, and taste fibers of the chorda tympani, to the anterior two-thirds of the tongue; CNs IX and X innervate taste buds of the posterior tongue and the base of the epiglottis, respectively. The facial nerve (CN VII) is the motor supply to the orbicularis oris. General sensation to the buccal mucosa is via the second division of the trigeminal nerve (maxillary nerve, CN V₂).

Physiology

On average, 1,500 mL of saliva is produced daily from the parotid, sublingual, submandibular, and minor salivary glands.

Detailed swallowing physiology is beyond the scope of this handbook. Briefly, swallowing is divided into active and passive phases. The active phases include a preparatory phase that involves salivation and mastication, and a second oral phase that involves bolus propulsion posteriorly. In the passive phase, CNs IX and X control involuntary laryngeal protective mechanisms and peristalsis.

◆ Pharynx

General

The pharynx extends from the skull base to the sixth cervical vertebra (C6) and is divided into the nasopharynx, superior to the palate; the oropharynx, extending from the palate to the hyoid and from the circumvallate papillae anteriorly; and the hypopharynx, inferior to the hyoid, including the piriform recesses, posterior wall, and postcricoid region (**Fig. 5.1**).

The cervical esophagus extends inferiorly, and the laryngotracheal complex sits anteromedially. Waldeyer's ring of lymphoid tissue includes the adenoids (pharyngeal tonsil) of the nasopharynx, the palatine tonsils of the oropharynx, and the lingual tonsil lining the base of the tongue. Taste buds and minor salivary glands exist in this region as well. The auditory tubes (eustachian tubes) open in the lateral nasopharynx.

Musculature

The superior, middle, and inferior pharyngeal constrictors surround the pharynx, enveloped by the visceral layer of cervical fascia. The palatopharyngeus and stylopharyngeus are supportive. The palatoglossus and palatopharyngeus form the tonsillar pillars.

Blood Supply

The lingual artery supplies the tongue. The palatine tonsils are supplied by external carotid branches via the facial artery, lingual artery, lesser palatine artery, descending palatine artery, and ascending pharyngeal artery.

Lymphatic Drainage

Rich bilateral drainage supplies the base of the tongue and piriform recesses and drains to levels 2 through 4. The tonsils drain primarily to the jugulodigastric region.

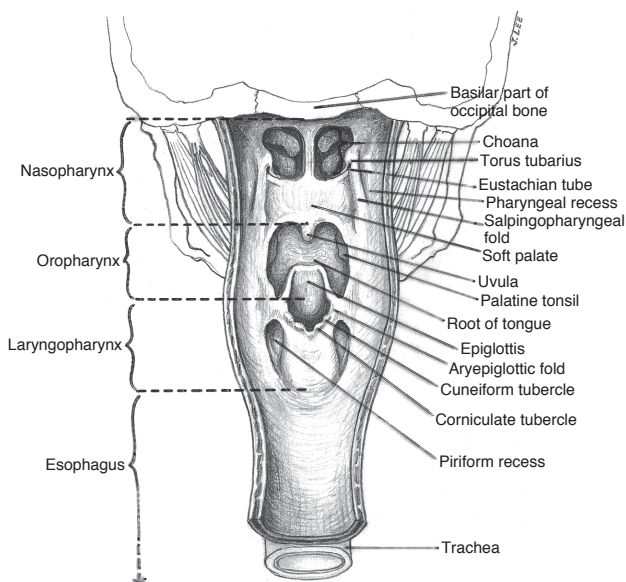


Fig. 5.1 Opened posterior view of the pharynx, demonstrating the boundaries of the nasopharynx, oropharynx, and hypopharynx. (Used with permission from Van de Water TR, Staecher H. *Otolaryngology: Basic Science and Clinical Review*. New York: Thieme; 2006:553.)

Nerve Supply

The palatine tonsils have sensory supply from the glossopharyngeal nerve (CN IX) and the lesser palatine nerve (branch of the maxillary nerve, CN V₂). Referred otalgia is common. CNs IX and X supply motor and sensory innervation to the hypopharynx.

Physiology

Swallowing is discussed with the physiology of the oral cavity, above.

♦ Larynx

General

The larynx can be considered to be a complex valve that regulates airflow. It is a dynamic organ that is involved with both the respiratory/vocal system and the digestive tract because of its position in the pharynx. Its lumen continues superiorly with the pharynx and inferiorly with the trachea; posteroinferiorly it is separated from the pharyngoesophageal lumen. The larynx is divided into the supraglottis, which includes the epiglottis, arytenoids, aryepiglottic fold, false vocal fold, and ventricle; the glottis, which is 1 cm inferior to the laryngeal inlet and includes the true vocal folds; and the subglottis, which extends inferiorly to the inferior border of the cricoid cartilage ring (**Fig. 5.2**).

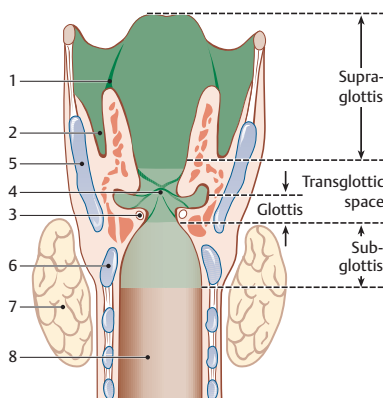


Fig. 5.2 Compartments and individual structures in the larynx. 1. The aryepiglottic fold, forming the boundary between the larynx and hypopharynx. 2. The piriform recess, which belongs to the hypopharynx. 3. Vocal ligament. 4. Anterior commissure. 5. Thyroid cartilage. 6. Cricoid cartilage. 7. Thyroid gland. 8. Trachea. (Used with permission from Behrbohm H et al. *Ear, Nose, and Throat Diseases: With Head and Neck Surgery*, 3rd ed. New York: Thieme; 2009:293.)

Skeleton

Three unpaired cartilages form the main laryngeal structure (**Fig. 5.3**). These are the epiglottis, the thyroid cartilage (from the Greek *thyreos*, meaning oblong shield), and the cricoid (from the Greek *krikos*, meaning ring). Three paired cartilages constitute the remainder of the laryngeal skeleton: the arytenoids, the corniculates, and the cuneiforms. Anterosuperiorly, the larynx is connected to the hyoid bone by the thyrohyoid membrane and muscle, and inferiorly it joins the trachea. Posteriorly, the larynx meets the muscular wall of the pharynx, with the cervical vertebrae posterior to this layer. The thyroid and cricoid cartilages are hyaline cartilage, which may ossify with age. The inferior horns of the thyroid cartilage articulate with the cricoid cartilage; the paired arytenoids articulate with the cranial border of the cricoid lamina. Both of these articulations are synovial joints.

Soft Tissue

Externally, the important membranes include the thyrohyoid membrane, the cricothyroid membrane, and the cricotracheal ligament. Internally, the membranous lining of the larynx is the quadrangular membrane superiorly, extending to the vestibular fold or false vocal fold, and the cricovocal membrane or conus elasticus, extending from the true vocal fold inferiorly. Paired aryepiglottic folds define the opening into the laryngeal lumen superiorly. Lateral and inferior are the piriform recesses, which funnel food and liquid into the esophagus. The paired vocal folds extend from the vocal process of the arytenoids dorsally to the thyroid cartilage ventrally at the anterior commissure. The structure of the vocal folds includes the vocal ligament, lateral cricothyroid ligament, median cricothyroid ligament, the vocalis muscle (thyroarytenoid), and the mucosal covering.

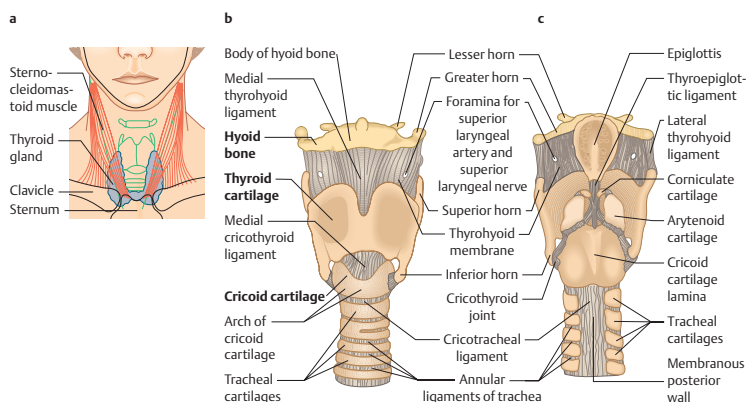


Fig. 5.3 Anatomy of the laryngeal bones and cartilage. (Used with permission from Probst R, Grevers G, Iro H. *Basic Otorhinolaryngology: A Step-by-Step Learning Guide*. New York: Thieme; 2006:338.)

Musculature

The posterior cricoarytenoid exclusively opens (abducts) the vocal folds. The lateral cricoarytenoid and transverse arytenoid (interarytenoid) closes the vocal folds, along with the thyroarytenoid. The thyroarytenoid and the cricothyroid place the vocal folds under tension.

Blood Supply

Arterial supply to the supraglottis arises from the external carotid via the superior laryngeal artery. The inferior laryngeal artery, arising from the subclavian artery via the thyrocervical trunk, supplies the subglottis. Venous drainage is to the internal jugular and brachiocephalic veins.

Lymphatic Drainage

The supraglottis has rich bilateral lymphatic drainage, with connections via the preepiglottic space. The glottis has few lymphatics. The subglottis also has rich bilateral drainage via paratracheal and pretracheal channels. The lymphatic drainage influences the frequency of metastatic spread of laryngeal carcinomas based on the sites of involvement of the primary tumor.

Nerve Supply

The vagus nerve (CN X) supplies the larynx. The superior laryngeal nerve has an external branch, providing motor function to the cricothyroid muscle, and an internal branch, providing sensation to the supraglottis and glottis. The recurrent laryngeal nerve provides motor supply to all other internal laryngeal muscles. On the left, the recurrent nerve passes around the aortic arch; on the right, it passes around the subclavian artery. Both recurrent nerves then ascend along the tracheoesophageal groove to enter the larynx at the inferior cornu of the thyroid cartilage. Importantly, the recurrent laryngeal nerve may branch in the neck prior to entering the larynx. (A

nonrecurrent right laryngeal nerve may occur if an aberrant subclavian artery is present.)

Physiology

Briefly, the basic functions of the larynx include airway protection, speech, and respiration. The larynx acts as a sphincter, in concert with pharyngeal structures, to prevent airway aspiration. This is facilitated via epiglottic tilt and contraction of the aryepiglottic folds, false vocal folds, true vocal folds, and adductors. Detailed discussion of the physiology of phonation is beyond the scope of this book.

♦ Esophagus

General

The esophagus lies posterior to the trachea. The esophageal opening is ~15 cm from the upper incisors at the lower border of the cricoid (C6), and the gastroesophageal junction is ~40 cm in adults. The upper esophageal sphincter is formed by the cricopharyngeus muscle at the esophageal opening. A middle physiologic sphincter exists at ~25 cm due to the aortic arch and left bronchus. The lower esophageal sphincter is at the gastroesophageal junction. The esophageal wall has a mucosa, submucosa, muscle layer, and outer fibrous adventitia. There are external longitudinal muscle fibers and internal circular fibers. The superior third is striated muscle, the middle third is mixed, and the inferior third is smooth muscle. The mucosa is stratified squamous, with a transition to columnar epithelium at the junction with the stomach.

Blood Supply

Superiorly, supply is via branches from the inferior thyroid artery originating from the thyrocervical trunk. In the thorax, supply is segmental via branches of the thoracic aorta or via intercostal vessels. Inferiorly, there are branches from the left gastric artery. There is a venous plexus, and drainage parallels supply generally.

Lymphatic Drainage

The cervical lymphatic system includes the deep cervical and paratracheal nodes; the thoracic lymphatics are composed of the posterior mediastinal and tracheobronchial nodes. The abdominal lymphatic system incorporates the left gastric and celiac nodes.

Nerve Supply

The esophagus is served by the glossopharyngeal and vagus nerves and the sympathetics. The myenteric plexus of Auerbach lies between the longitudinal and circular muscle layers.

Physiology

The esophageal phase of swallowing occurs as the bolus passes the upper esophageal sphincter. Primary peristalsis has an initial rapid inhibitory

phase followed by a longer wave of contraction. The lower esophageal sphincter relaxes to open for 6 to 8 seconds. There is also a secondary peristaltic wave in a rostral-to-caudal progression.

5.1 Laryngeal and Esophageal Emergencies

5.1.1 Stridor

◆ Key Features

- Stridor is an indication of airway compromise and should be considered an emergency.
- The priorities are to identify the site of obstruction, restore adequate ventilation, and address the underlying cause.

Stridor is an exam finding defined broadly as noisy breathing due to partial upper airway obstruction, and it is usually high-pitched and harsh. This is to be distinguished from wheezing, which is noise due to reversible collapse of bronchioles of the lower pulmonary airway, and from stertor, a sonorous noise that is due to collapse or obstruction at the upper pharynx, such as snoring.

◆ Clinical

The stridulous patient must be evaluated without delay, as loss of airway may progress rapidly.

Signs and Symptoms

In the adult, development of stridor may be acute or chronic. Symptoms are often nonspecific and variable. Associated symptoms will vary depending on etiology and may include dyspnea, pharyngodynia, dysphagia, odynophagia, and anxiety. Signs include audible noise with breathing and may include fever, cough, hemoptysis, and retractions. Depending on severity and acuity, the patient may be in distress, may be hypoxic, and may also display dysphonia.

Differential Diagnosis

The causes of stridor are numerous. It is useful to categorize based on the anatomic level of obstruction. As a generalization, inspiratory stridor correlates with supraglottic obstruction, expiratory stridor correlates with intrathoracic obstruction (trachea), and biphasic stridor suggests glottic or subglottic obstruction (**Table 5.1**). Differential diagnosis includes an obstructing neoplasm, infection with edema (e.g., supraglottitis), allergy/

Table 5.1 Stridor localization by anatomic site

	Retractions	Stridor*	Voice	Feeding
Naso/oropharynx	Minimal	Stertor [†]	Normal	Normal [†]
Supraglottis	Marked and severe	Inspiratory and high-pitched	Muffled	Abnormal
Glottis/subglottis	Mild to severe	Biphasic and intermediate-pitched	Normal to very abnormal (barking cough)	Normal
Intrathoracic trachea	Mild to severe	Expiratory and low-pitched	Normal (seal-like cough)	Normal

*The quality of the airway noise.

[†]Unless associated with complete nasal obstruction in a neonate.

(Adapted with permission from Van de Water TR, Staecker H. *Otolaryngology: Basic Science and Clinical Review*. New York: Thieme; 2006:214.)

angioedema, foreign body, traumatic injury to the airway (e.g., thyroid cartilage fracture, large hematoma), and bilateral vocal fold immobility (e.g., postoperative iatrogenic injury). Other causes include subglottic stenosis, tracheal stenosis, and tracheomalacia.

◆ Evaluation

One must consider an acutely stridulous patient as a potential airway emergency; prompt evaluation is warranted.

Physical Exam

In critical care algorithms, the ABCs are followed: airway, breathing, circulation. Stridor is a reflection of airway narrowing and is therefore a priority. If the stridulous patient is not ventilating adequately (hypoxic or retaining CO₂), rapid intervention is critical. Establishment of the airway is discussed subsequently under Treatment Options.

History is important in guiding the exam: the timing of onset; known diagnoses, such as history of angioedema or head and neck cancers; previous head and neck surgeries (thyroid surgery, previous tracheotomy); trauma; possible foreign body aspiration; current upper respiratory infection; history of intubations; and any other relevant facts.

On exam, vital signs; pulse oximetry; possibly arterial blood gases; phonation; an oral and pharyngeal exam; and a neck exam for masses, edema, crepitus, or tenderness are important. Unless the adult patient is unstable or not adequately ventilating, a flexible fiberoptic nasopharyngolaryngoscopy is usually safe and extremely helpful. This exam will reveal an estimation of glottic airway diameter, vocal fold mobility, any sites of edema or mass, or the presence of an obstructing laryngeal foreign body. Pediatric patients may selectively undergo flexible fiberoptic examination. Caution must be used, as the examination can precipitate further airway compromise.

Imaging

Plain films of the neck—posteroanterior (PA) and lateral views—may be informative but are being replaced by computed tomography (CT) or magnetic resonance imaging (MRI). In general, one should not send a patient with airway compromise for CT or MRI unless the airway has been secured; loss of the airway during imaging can be disastrous.

Labs

An arterial blood gas (ABG) test may be helpful in determining adequacy of ventilation. Less acutely, a complete blood count (CBC) with differential may be useful in a patient with infection, such as supraglottitis.

◆ Treatment Options

The goals of treatment are (1) to determine the site(s) and degree of obstruction; (2) to stabilize the airway by forced ventilation, intubation, or surgical bypass of the site of obstruction; and (3) to treat the underlying cause.

In an unstable, poorly ventilating patient, the airway must be secured. If possible, the operating room is the safest place to do this. One should approach the airway problem algorithmically, thinking ahead about possible problems (with a plan B and plan C). The algorithm will differ depending on the etiology and the patient. There are useful generalizations, however. (1) It is absolutely critical that the otolaryngologist clearly and authoritatively take control of the airway management. (2) An upright, awake, spontaneously breathing patient is usually the safest situation. (3) One must be prepared for the possibility that sedating and paralyzing a stridulous patient for intubation may result in an inability to mask ventilate, precipitating an emergency. (4) With angioedema, tongue swelling usually precludes the ability to perform direct laryngoscopy.

If intubation is deemed to be impractical (e.g., fiberoptic laryngoscopy indicates that an endotracheal tube will not pass through a narrowed airway), awake tracheotomy is performed. In an emergency where the surgical airway must be most rapidly established, cricothyroidotomy is indicated.

Unless the fiberoptic laryngoscopy suggests otherwise, an awake fiberoptic nasotracheal intubation is often the procedure of choice (if the patient requires intubation). As a backup plan, one should have a Holinger laryngoscope, velvet-eye laryngeal suction, and Eschmann stylet assembled and ready to use. Often, the otolaryngologist can easily intubate a patient with these instruments. A ventilating rigid bronchoscope is also very helpful, if available. In these cases, the patient should be maintained with spontaneous ventilation; if the patient has airway masses or stenosis, then ventilating bronchoscopy can be diagnostic and therapeutic. In addition, a tracheotomy tray should be open and ready to use. Injecting the soft tissue over the cricothyroid membrane with 1% lidocaine and 1:100,000 epinephrine ahead of time will result in vasoconstriction and a much drier operative field if emergency cricothyroidotomy or tracheotomy becomes necessary.

Other strategies for difficult intubation include retrograde intubation by placing a needle and guidewire (from a central line kit) into the cricothyroid membrane or trachea and passing the guidewire up and out of the mouth.

An orotracheal tube may then be blindly passed over the guidewire and into the trachea. There are other techniques, such as fiberoptic intubation through a laryngeal mask airway, video direct laryngoscopy, intubating endoscopes (e.g., Shikani optical stylet), or the use of a lighted stylet to be blindly introduced into the trachea.

Medically, there are helpful strategies to “buy time” or assess response to medical therapy if a patient can maintain ventilation. The patient is maintained in an intensive care unit with continuous pulse oximetry monitoring. Humidified oxygen (e.g., via face tent) will help minimize secretions. Heliox (typically 79% helium/21% oxygen mixture) has been advocated as a short-term intervention to help maximize ventilation while definitive intervention is planned. The gas functions by reducing the viscosity of the inspired air, thus reducing the mechanical work of breathing in the narrowed airway. It can be used while medical intervention is taking effect; this is an excellent means of avoiding intubation. Intravenous (IV) steroids (i.e., dexamethasone 8 mg IV every 8 hours) may reduce airway edema. In some situations, appropriate medical treatment of the underlying problem, such as infection or angioedema, can obviate the need for intubation or surgical airway.

◆ Outcome and Follow-Up

After securing the airway, appropriate management directed at the underlying problem is undertaken. This may include biopsy, treatment of infection, and laboratory or radiographic work-up.

5.1.2 Laryngeal Fractures

◆ Key Features

- Airway obstruction can develop rapidly.
- Evaluate for concurrent injuries such as pneumothorax or esophageal or vascular injury.
- The treatment goals are to ensure an adequate airway, to maintain voice quality, and prevent aspiration.

Laryngeal fractures can be mild, nondisplaced, and isolated. They may also be severe and displaced, with airway compromise and concomitant injuries to other structures of the head, neck, and chest. A fiberoptic laryngeal exam and a computed tomography (CT) scan will usually direct care appropriately.

◆ Epidemiology

Five to ten percent of traumatic injuries involve the neck. Penetrating neck injuries are more common than blunt trauma (see also **Chapter 6.1.4**).

◆ Clinical

Patients usually present with a history of blunt trauma to the anterior neck. Common mechanisms include assault, a strangling or hanging attempt, and vehicular accidents including automobiles, snowmobiles, motorcycles, all-terrain vehicles, and bicycles. A gunshot wound to the neck can result in a cricoid or thyroid fracture alone or in combination with other injury.

Signs and Symptoms

Signs and symptoms may include dysphonia, aphonia, stridor, cough, hemoptysis, dysphagia, and pain. A hallmark of severe injury is subcutaneous emphysema, as well as loss of the normal prominence of the thyroid cartilage.

Differential Diagnosis

Differential diagnosis is based on severity. The injury may be confined to soft tissue or may include cartilage fracture, mucosal disruption, or additional injuries to the pharynx, esophagus, or major vessels.

◆ Evaluation

Physical Exam

A fiberoptic exam and CT are diagnostic. The injury may be occult. A delay in diagnosis contributes to mortality. Trauma algorithms are followed, including ABCs (airway, breathing, circulation). In the presence of laryngotracheal cartilage disruption, intubation is generally contraindicated. An emergency airway is established via a tracheotomy. Transtracheal jet ventilation can be extremely useful while the definitive airway is being established. The remainder of the head and neck exam is directed at determining additional injuries. With a questionable laryngeal fracture, flexible fiberoptic laryngoscopy is performed. The presence of obvious endolaryngeal mucosal disruption, bleeding, or cartilage derangement suggests the injury.

Labs

Laboratory tests are not routinely indicated. When following an inpatient after a laryngeal fracture, a complete blood count (CBC) with differential may be useful to monitor for infection and the development of mediastinitis or sepsis.

Other Tests

Once the airway is deemed stable or stabilized, imaging should be obtained. A CT scan of the neck, viewed with bone windows, provides excellent detail of the laryngeal framework. A chest X-ray is mandatory to evaluate for pneumothorax. A cervical esophagogram is recommended to evaluate for occult esophageal injury.

◆ Treatment Options

The acute treatment goal is maintaining or establishing the airway, as just discussed. The long-term treatment goals include maintaining an adequate airway, a satisfactory voice, and the ability to swallow without aspiration.

Larynx trauma can be categorized as mild or severe. Mild injuries include laryngeal edema, simple endolaryngeal lacerations with minimal cartilage exposure, and nondisplaced fractures. If severity is unclear, an examination under anesthesia with direct laryngoscopy and esophagoscopy is recommended. Patients with mild injuries can usually be managed non-operatively. Treatment must include observation in a monitored bed (i.e., continuous pulse oximetry), along with intravenous (IV) steroids (Decadron [Merck & Co., Inc., Whitehouse Station, NJ] 8 mg IV every 8 hours $\times$ 3 doses), humidified air/O₂, and a soft diet.

Patients with severe mucosal injuries and/or displaced or comminuted cartilage fractures are managed surgically with low tracheotomy followed by exploration and repair. Fractures should be repaired early—ideally, within 24 hours. The basic goals include closure of tracheal defects, coverage of exposed cartilage, removal of devitalized cartilage, and reduction of cartilage fractures. After tracheotomy, direct laryngoscopy and esophagoscopy are performed. If the vocal folds are significantly displaced, laryngofissure is performed with repair of mucosa. Fractures are reduced and reapproximated with Prolene (Ethicon, Somerville, NJ) suture, wire, mesh, or plating. If possible, external perichondrium should be closed. Stenting may be necessary to prevent stenosis or anterior webbing. Indications for stenting are an area of controversy.

◆ Outcome and Follow-Up

Posttreatment follow-up includes ongoing assessment of the airway and swallowing. A barium swallow study with speech-language pathology for evidence of aspiration is important. Consideration of tracheotomy decannulation is based on standard criteria. Further surgical management of webbing or stenosis may be required, as well as long-term use of a T-tube. Patients may require voice therapy.

5.1.3 Caustic Ingestion

◆ Key Features

- Perform an airway assessment and intubation if there is any compromise.
- Identify the ingested substance.
- Do an early endoscopy.

◆ Epidemiology

The incidence of caustic ingestion is estimated at 5,000 to 15,000 cases annually in the United States: 53% of cases occur in children 6 years of age or less. However, only ~3% of deaths secondary to caustic ingestion occur in young children.

◆ Clinical

The patient with caustic ingestion presents with varying findings depending on the type of substance ingested and the quantity.

Signs and Symptoms

With moderate to severe injury, symptoms include oral, neck, and chest pain and dysphagia. Young patients may drool, and older patients may spit secretions or refuse to swallow. Respiratory problems range from coughing and wheezing to stridor and respiratory distress.

Differential Diagnosis

The injury may range from mild to severe as a result of acid versus alkaline ingestion.

◆ Evaluation

It is important to obtain information about the substance ingested. Alkaline substances (lye, ammonia, soaps) cause liquefaction necrosis and deep penetrating injuries to the pharyngeal and esophageal wall. Ingestion of lye (sodium hydroxide, drain cleaner) accounts for ~60% of cases. Acids cause a mucosal surface coagulative necrosis, and deep tissue injury is uncommon. Household bleach is usually less than 5.25% sodium hypochlorite and, therefore, usually causes only mild mucosal irritation.

Physical Exam

Assessment of the airway is the first priority. If there is evidence of airway compromise, the airway should be secured by intubation before edema worsens. This is especially important in children. A head and neck exam and skin survey for caustic burns from the spilled substance are indicated, followed by assessment of the oral and pharyngeal mucosa for injury. Blind nasogastric tube placement is avoided; emesis is not induced; charcoal gastric lavage is not indicated. In the past, the timing of endoscopy was controversial. Currently, esophagoscopy within 24 to 48 hours is recommended. Damage, in terms of an estimate of the depth of injury, is assessed at the proximal extent of injury. Additional advancement of the rigid endoscope is not performed, to avoid further injury.

Labs

Routine preoperative labs are obtained if surgery is planned. A complete blood count (CBC) is followed if there is suspicion of mediastinitis.

Table 5.2 Grading of esophageal caustic burns by endoscopic findings

1st degree	Erythema and punctuate hemorrhages Superficial mucosal injury
2nd degree	Ulceration, exudates, pseudomembranes Partial-thickness injury
3rd degree	Eschar, perforation into or through muscular layer Full-thickness injury

Other Tests

Acutely, a contrast swallow study is not routinely indicated. However, contrast swallow studies are often used to assess the progression of stricture formation during recovery.

◆ Treatment Options

Nonoperative therapy includes hydration, acid-blocking medications, and observation. Steroids do not seem to reduce the incidence of stricture formation; some authors have suggested that they may mask the signs or symptoms of perforation. Antibiotics are often given empirically until perforation has been ruled out. An endoscopy is performed to grade the injury (**Table 5.2**). In general, the scope should not be passed beyond any segments of necrosis or concern for transmural injury. A thoracic surgery consultation is obtained for management of perforation or high-grade injuries. There has been a trend toward early surgical intervention by thoracic surgeons (within 36 hours) to treat deep esophageal injuries. When there is evidence of a high-grade injury on endoscopy, a feeding tube should be placed under direct vision.

◆ Outcome and Follow-Up

Mortality ranges from 0 to 8% overall. Complications including mediastinitis, perforation, and stricture formation range from 10 to 20%. Stricture development generally begins at week 3 or 4; a contrast swallow study or a follow-up endoscopy is helpful at this time.

5.1.4 Laryngeal Infections

◆ Key Features

- Infections of the larynx may be acute or chronic in nature.
- Symptoms and signs include hoarseness, pain, dysphagia, cough, dyspnea, stridor, and signs of systemic illness.
- Therapy is directed at the causative organism and is supportive of any airway issues present.
- Laryngoscopy with biopsy and/or culture may be necessary, as some infections may mimic neoplasms, or required for a definitive diagnosis.

“Laryngitis” is a term used to describe any inflammatory condition of the larynx. Infectious laryngitis may be caused by viral, bacterial, mycobacterial, fungal, and even protozoan organisms. History and physical examination, including laryngoscopy, are instrumental in establishing the diagnosis. The immunologic state of the patient, travel history, and exposure to illness are important factors to address in the history.

◆ Epidemiology

Viral laryngitis is the most common type of laryngitis. Incidence is difficult to establish, as most patients do not seek out care for acute laryngitis issues. Secondary to airway size, laryngitis may present in children with airway symptoms; adults more commonly present with hoarseness and pain complaints.

◆ Clinical

Signs and Symptoms

Viral Laryngitis

Viral laryngitis usually accompanies a viral upper respiratory infection (URI) with systemic viral syndrome. Symptoms include hoarseness, with pitch breaks, and decreased pitch; cough; and pain. Examination shows erythema and edema of the vocal folds. Agents include those causing URIs, with rhinovirus being most common. Treatment is symptomatic, as this should be a self-limited disease. Therapy includes voice rest, hydration, humidification, cough suppressants, and expectorants.

Croup

Croup (laryngotracheitis) is a viral infection seen in children characterized by stridor, a “barky” cough, and fever. This illness is seen primarily in winter but may occur year-round. The severity of the illness varies widely and is based on the degree of subglottic edema. Parainfluenza viruses 1 and 2 and influenza A are the most common etiologic agents. The most important assessment of these patients is evaluating the respiratory

status for impending airway intervention and support. Initial home management involves humidification or steam shower, although this is not evidence-based. Racemic epinephrine may be useful to those children with respiratory distress, serving to decrease airway edema rapidly. Admission is no longer thought to be absolutely necessary after this treatment, as the “rebound” concern is rare. Corticosteroids have a clear role in treatment of this condition. Airway intervention including intubation or tracheotomy may be necessary if respiratory decompensation occurs, with respiratory fatigue, hypercarbia, inadequate oxygenation, or worsening neurologic status. For intubated children, an air leak should develop and indicate extubation potential, usually within 2 or 3 days.

Bacterial Laryngitis

Bacterial laryngitis is far less common than viral etiologies. Supraglottitis may involve the entire supraglottis or, more focally, the epiglottis (epiglottitis). The incidence of epiglottitis in children has dramatically fallen since the introduction of the *Haemophilus influenzae* type B vaccine. Adults typically have more diffuse supraglottitis. Patients relate sore throat, fever, odynophagia, muffled voice, and dyspnea. Children may have drooling. Patients may sit in a tripod position, with the torso leaning partially forward, arms positioned at the sides just in front of the torso, and neck extended. Diagnosis is confirmed by the swollen, erythematous epiglottis or supraglottis seen on flexible laryngoscopy. In children, the diagnosis may best be made by lateral neck film and a “thumbprint” sign representing a swollen epiglottis, as manipulation of the airway with flexible endoscopy may precipitate airway compromise. Although historically *H. influenzae* is the most common organism in children, other organisms such as *Streptococcus pneumoniae*, *Staphylococcus aureus*, β -hemolytic *Streptococcus*, *Klebsiella pneumoniae*, and parainfluenza virus may be etiologic agents. In severe cases, a secure airway should be established, potentially in an operating room setting. Less emergent airways may be managed with hospital admission and close observation. Adults are less frequently in need of airway interventions than children are. Treatment should include antibiotics that cover β -lactamase-producing *H. influenzae*, such as a second- or third-generation cephalosporin, as well as corticosteroids. Epiglottitis abscess is a rare complication of supraglottitis.

Fungal Laryngitis

Although fungal infections occur more commonly in the immunocompromised patient, fungal laryngitis may be seen in many varied patient populations. These infections may present as leukoplakia, either focal or diffuse. Risk factors for fungal laryngitis include those with diminished systemic immune response (e.g., immunocompromised states, diabetes, human immunodeficiency virus [HIV], immunomodulating medications, and severe nutritional deficiencies), or depressed local immune function (e.g., prior local radiation therapy, corticosteroid inhalers, smokers). *Candida* sp. are the most common organisms, but blastomycoses, *Histoplasma*, *Aspergillus*, *Cryptococcus*, and coccidiomycosis may be involved. Secondary to the infrequency of these infections and their presentation as leukoplakia,

biopsy may be done. Some infections can show pseudoepitheliomatous changes and can even be mistaken for squamous cell carcinoma. Special fungal stains may be necessary. Systemic antifungal agents appropriate for the causative fungal organism are necessary. Topical antifungal medications are not curative, as they require direct contact with the organism, and the larynx is protected from exposure through protective-airway mechanisms.

Tuberculosis

Tuberculosis of the larynx may be seen with pulmonary infection (historically one of the most common laryngeal infections), but may be experienced as an isolated infection in 20 to 40%. Risk factors include exposure in endemic areas, immunocompromised states, and nursing home environments. Patients typically report hoarseness, dysphagia, odynophagia, and cough. The larynx typically is edematous and hyperemic in its posterior third, with some exophytic granular areas. Biopsy, chest radiography, and sputum cultures may be necessary for diagnosis. Treatment is with antimycobacterial drugs, with cultures helpful given the significant drug resistance that may be present.

Actinomycosis

Actinomycosis is an unusual laryngeal infection; it is more commonly known to affect the oral cavity. The fastidious anaerobic gram-positive bacterium has some features resembling fungi. Infections may manifest as edema, abscess, and ulceration. Biopsy may demonstrate the pathognomonic sulfur granules. Treatment typically is with high-dose penicillin given intravenously (IV) or orally depending on the disease severity. Clindamycin is an option for penicillin-allergic patients.

Syphilis

Luetic (syphilitic) infections were once much more common than today. Syphilis may be present in the primary, secondary, or tertiary stage. Laryngeal involvement is rare and may be a component of the secondary or tertiary (gumma) phase. Caused by the *Treponema pallidum* spirochete, these infections may lie dormant for prolonged periods of time. Laryngeal involvement may present as a diffuse hyperemia, ulceration, or maculopapular mucosal rash. Laryngeal syphilis may cause fibrosis, chondritis, and scarring or stenosis. Serologic testing should include Venereal Disease Research Laboratory (VDRL) testing (100% sensitivity in secondary syphilis), with confirmation using fluorescent treponemal antibody absorption (FTA-ABS) and the microhemagglutination assay for *T. pallidum* (MHA-TP). Penicillin G is the primary drug in treatment.

Leprosy

Leprosy, or Hansen's disease, caused by *Mycobacterium leprae*, is very rare in the United States but is seen more often in the Indian subcontinent and Africa. Nasal infection is the primary site of infection in the head and neck, with the larynx being second. Infection favors the supraglottic, with symptoms including hoarseness, muffled voice, odynophagia, and cough. Examination may reveal erythema, nodularity of the mucosa, and ulceration.

However, often the patient has minimal pain, despite the appearance of the tissue. Scarring may develop that leads to stenosis and airway obstruction. Biopsy may reveal “foam cells,” with copious *M. leprae* within these cells. Long-term medical therapy is necessary, even 5 to 10 years, with dapsone and rifampin.

Other

The larynx may be involved in other systemic infections, particularly viral infections such as mumps, measles, or varicella (chickenpox).

Differential Diagnosis

There are several causes of symptoms similar to those of laryngeal infection:

- Laryngeal neoplasm
- Airway foreign body
- Autoimmune diseases
- Allergic reaction
- Angioneurotic edema
- Subglottic stenosis
- Laryngopharyngeal reflux

◆ Evaluation

History

The key in evaluation is primarily history, including time course of the condition, exposures, comorbidities, and past medical history.

Physical Exam

A complete, thorough head and neck examination, including flexible nasolaryngoscopy, should be performed. The lungs should also be auscultated to evaluate for concomitant pulmonary issues.

Imaging

The role of imaging is limited. Computed tomography (CT) may be helpful if an abscess is suspected or to aid in evaluating the complication of airway stenosis. If tuberculosis is suspected, chest radiography should be ordered.

Labs

Laboratory testing may include a complete blood count (CBC) or specific serology as appropriate. In some cases (e.g., epiglottitis), laboratory testing would be contraindicated initially. Tissue biopsy and culture may be necessary to confirm a diagnosis for several reasons. First, many of these infections are uncommon; therefore, clinical experience is limited in recognizing the entity definitively. Second, the laryngeal findings may mimic

squamous cell carcinoma grossly; the onus is on the clinician to evaluate for this issue. Last, given that the medical therapy may be prolonged and not without side effects and medication interactions, a tissue culture assists in quality care.

◆ Treatment Options

Treatment of laryngeal infections is based on the causative agent. Viral laryngitis is managed with supportive measures. Pharmacologic treatment should be tailored to the organism implicated in the infection.

◆ Outcome and Follow-Up

Follow-up with laryngoscopy should be used to assess resolution of the infection. The frequency of these evaluations is determined by the severity of the infection and the expected time to resolution.

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